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Adaptive few-shot learning with a fair priori distribution ☆, ☆☆

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ABSTRACT

Few-shot learning is a special challenge in pattern recognition, which identifies unseen categories given only limited samples. In the past few years, various methods have been proposed to solve few-shot problems. Based on the predecessors, we discuss the prior influence of the training set bring into the model, the relationship between data quality and feature performance, as well as the method and reason to expand negative samples. We consequently improve the models and learning strategies in these aspects, then we obtain a novel framework named AFSL: Adaptive Few-Shot Learning, which can eliminate the impact of data sets on the model, and also performs well on several public datasets. Specifically, we achieved a percentage increase of verification accuracy of 7.8% and 1.5% for 1-shot and 5-shots tasks comparing to prototype network on mini-Imagenet dataset, 9.8% and 2.4% on tiered-Imagenet, and 28.4% and 14.0% on the CUB dataset.

1. Introduction

Nowadays, the deep learning method has made great achievements in various fields, but the performance of deep learning methods heavily relied on a large scale of labeled data, otherwise, these models will inevitably lead to over-fitting in the case of only few samples accessible. The reality is that there is not enough labeled data in some specific field. Consequently, the *few-shot learning* problem emerges, aiming at training models with only few samples, And can identify unseen categories.

Some traditional and common methods such as regularization, fine-tuning and data augment are used to enhance classification results, However, the few-shot problem has its own characteristics not only including the lacking of data. Many algorithms were designed specifically for few-shot learning in recent years. We review a subset of these methods in Section 2. Although former researches significantly improved the performance of few-shot learning, there still exists essential drawbacks in these works. That is the estimation of data distribution based on limited data is also obviously biased. Furthermore, an easy but effective way of expanding the size of negative samples, in order to better constrain the model, still remains an open issue.

In this paper, an improved Adaptive Few-Shot Learning (AFSL) approach is proposed. The main contributions of our work include:

Firstly, a novel but simple logit prediction method is proposed in order to promote few-shot tasks. The method's decision surface varies for query samples with different qualities, which plays a certain role in hard-sample-mining, making the prediction more meaningful and effective.

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Secondly, Through theoretical analysis, our method can also remove the biased apriori introduced by the training data statistics, which sheds new light on the probabilistic consideration of few-shot learning.

Thirdly, we present a modified episodic learning algorithm. With advanced data sampling manner and contrastive constraints, the number of negative samples in the same batch are nearly doubled, which we claim as an essence of metric learning.

Last but not least, the proposed novelties are evaluated on various datasets. Results competitive to the state-of-the-art are shown and consequently, their effectiveness is proven.

The remains of this paper are organized as follows: In Section 2, examples of existing solutions related to the few-shot task are reviewed. Section 3 mainly introduces our improved methods, and also corresponding motivation and analysis are included. And in Section 4, a series of experiments are designed, whose results are shown to support the effectiveness of our assumptions and proposed approaches. Finally, a conclusion is summarized in Section 5.

2. Related works

In past few years, many methods have been proposed to improve few-shot learning. Discussing from different aspects, but they mainly based on the same principle: to extract more robust and discriminative features, and to find out an appropriate way modeling the similarities or distances among examples.

2.1. Episodic learning

In order to make the few-shot learning model adapt to new tasks quickly, a special way to organize data is needed. The ability of rapid transfer is achieved by learning from a series of different task combinations. Originally, this concept came from Model-Agnostic Meta-Learning (MAML) [1], its main idea is to find an optimal set of initialization parameters, which can be efficiently adjusted to fit new tasks within few steps of updating. Following it, On first-order meta-learning algorithms (reptile) [2] made further improvement by simplifying the derivation process, The key point is their training data sampling schedule. Briefly, query set and support set are sampled separately from a sub-label space of training data. Few-shot tasks are created as recalling each query into its corresponding category with limited support data in the same batch. Nowadays, it is also called the episodic learning [3]. The episodic training strategy is carried out repeating the following steps:

1. Randomly pick C classes from D_{train};
2. Sample K_s images from each class, which constitute the support set $D^S_{train} = \{(x_i, y_i)\}^m_{i=1}$, where $m = K_s * C$, S means the support;
3. From each of the same C classes, sample K_q images serve as the query set $D^Q_{train} = \{(x_j, y_j)\}^n_{j=1}$, where $n = K_q * C$ and $D^S_{train} \cap D^Q_{train} = \emptyset$, Q means the query;
4. Perform the C-way K_s-shot learning using D^S_{train} and D^Q_{train} to update the parameters.

C is used to represent the number of categories in a batch and K for the number of images in each category, the so-called C-way K-shot problem is typically a few-shot classification task. The dataset is divided into two separate parts D_{train} and D_{test}, whose label spaces are disjoint. In the subsequent researches mainly follow the above strategy, the same is true of the way in this paper.

2.2. Nearest neighbor metric

In order to enable the model to classify unprecedented categories, the nearest neighbor method is the most commonly used method. Matching network [4] (“matchingnet” for short hereafter) and prototype network [5] (“protonet” for short hereafter) raise an important conception. The “prototype” takes the feature mean of the support set as the same category is proposed. According to this “prototype”, the samples of the query set are classified by the nearest neighbor method. The matchingnet is similar to it, but the matchingnet measures cosine distance, while the protonet uses Euclidean distance. After Them, various methods emerge as endlessly, such as FGNN [6] based on Few-shot with Graph Neural Network (FGNN) structure and Few-shot learning via Embedding Adaptation with set-To-set functions (FEAT) [7] based on attention mechanism. But in fact, some methods focus on the metric function can also achieve great results. Quite a few metric learning methods followed the stream, and tricks like “margin” were further added into the cosine measurement, such as arcface [8], sphereface [9], cosface [10]. And some methods such as baseline++[11] and simpleshot [12], The baseline++ [11] uses a kind of weighted sum method to calculate the prototype’s center point, and simpleshot [12] measures Euclidean distance by subtracting all features’ mean value of base data and normalizing this feature. Through such simple changes in metric methods, they can still achieve a good performance. This hits the great potential for metric function. Following the previous work, after analyzing the characteristics of few-shot classification task, we also propose an improved method on metric function, and explain the rationality of our method.

2.3. Extend negative samples

In addition to the contents mentioned above, the number of negative samples in the same training batch is also a important factor for the improvement of model performance. According to Noise-Contrastive Estimation (NCE) [13] theory, the samples belonging to the same category can be regarded as positive samples, while samples of other categories can be regarded as negative samples. The more negative samples help to compress the within class distance, so as to enhance the discriminative capability. Therefore, how to design a sampling strategy to obtain more negative samples in one batch is regarded as a key point. In other words, these strategies are also named hard-sample-mining. Their goals are the same, the embedding features of images of the same categories should be mapped closer together in the feature's space than the embedding features of images of different classes. The Xuan and al [14] emphasizes the importance of hard negative samples again, and tries to analyze and solve the problem of model collapses caused by the hard negative in deep metric learning. And Robinson and al [15] discusses what kind of difficult samples are useful for the model. That is close to the sample's but belongs to a different class. Specifically, it mainly selects such samples in an unsupervised way. Along with this ideology, we also propose our strategy to expand the number of negative samples and hard samples mining.

3. The training framework

After matchingnet [4] and protonet [5], The following work basically follows this training framework. Mainly including the above-mentioned episodic learning and a nearest neighbor metric. In this paper, we have changed these two parts. In episodic learning, we designed a strategy that can double the number of negative samples in the same batch. In metric function, combined with the characteristics of few-shot task, we proposed an adaptive measurement method, which can be well combined with episodic learning to mine hard samples.

3.1. Improvement on metric function

The few-shot learning classification task, which is fundamentally a classification problem, is typically solved in the following paradigm: Firstly, d-dimensional feature $x \in \mathbb{R}^d$ of a given sample is extracted using some non-linear function f, which is usually a neural network nowadays. A typical set of hyperparameter d is 256 or 512, Then the feature x is compared with each category $y \in [0, C)$ in support set of a C-way problem, where each category takes the form as a "feature template" $w_y \in \mathbb{R}^d$; the inner-product operator is commonly utilized to calculate the "logit", the short form of "Logit Model", $l = \langle W, x \rangle = W^T x$, where $W \in \mathbb{R}^{d*C}$ is a weight matrix enumerating all categories' w_y, and the logit vector $l \in \mathbb{R}^C$. The logit is consequently fed into a frequently used *softmax* function $p_y = e^{l_y} / \sum_{i=1}^{C} e^{l_i}$, in order to be normalized into probabilities. Category in support set with the largest probability would be picked as the classification result.

Here we focus on the feature comparison, also known as the "logit prediction" step in the above paradigm. It is very straightforward to take Euclidean distance $D_{Eu} = \|w_y - x\|$ as the metric for comparison protonet [5] treat $-D_{Eu}$ or $-D_{Eu}^2$ as the logit, because it is negatively correlated to the probability of matching. It is worth noticing that a cosine means of comparison becomes popular in recent years, which takes $\cos\theta_y$ as the logit where θ_y the angle between two vectors x and w_y. The method is especially attractive since it is more efficient compared to Euclidean distance, while having competitive precision. Following this fashion, the feature and weight vectors are both mapped onto a unit hyper-sphere as follow:

$$\cos\theta_y = \frac{w_y^T x}{\|w_y\|\|x\|} = \frac{w_y^T}{\|w_y\|}\frac{x}{\|x\|} \tag{1}$$

But if you directly use the cosine metric, you cannot get a good result. This is because the output value of cosine is relatively small, and the absolute value is between (0, 1). The probabilities outputs from the softmax function do not differ much, even if a sample is correctly classified. Therefore, the s hyperparameter is added to amplify the cosine value, in the cosine formulas which played exactly the same role as "temperature" parameter in some other literature. The factor s was usually hand-crafted, for example a typical value 64, but sometimes it could also be learnable.

The cosine similarity can be easily deduced from (1) that magnitudes of both w_y and x are discarded, remaining the angular information only be such a simplification perfectly appropriate? In our opinion, the answer is probably NO, and we propose a simple but effective solution as follow:

$$l_{AFSL} = \frac{w_y^T x}{\|w_y\|} = \|x\|\cos\theta_y \tag{2}$$

The main structure of this model in this paper can be shown as Fig. 1, after data sampling, the samples are divided into support set and query set. Then, putting them into the backbone to obtain the corresponding feature output. The features of the support set as the prototype are normalized, but the features from query set are not, such operation is shown in (2). Finally, the results after AFSL are put into softmax function for obtaining the classification probability.

At first glance, it is almost simply replacing the re-scale factor s with feature L2 norm $\|x\|$. Since $\|x\|$ is partially determined by the backbone f, it is flexible in training, performing like a trainable s to solve the cosine range issue. But we argue, this is just a side effect, we describe details from two aspects, by distinguishing the above logit prediction function (2) from its two homologous relatives, namely the inner-product and cosine manners.

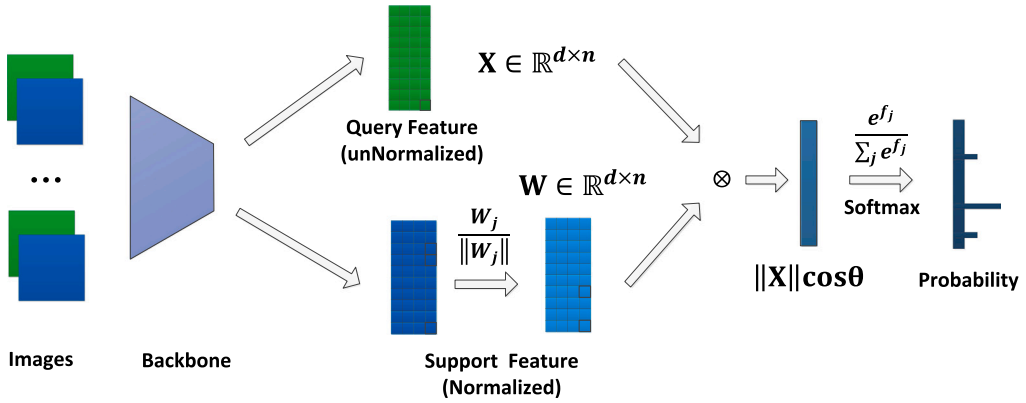


Fig. 1. Schematic diagram of network structure.

Differ from inner-product method: Recalling the basic classification paradigm, where we consider the inner-product operator taking following format:

$$l_y = \langle w_y, x \rangle = w_y^T x = \|w_y\| \|x\| cos\theta_y \tag{3}$$

logit l_y is then fed into the monotonic softmax function, hereby a normalized probability p_y is obtained, and it is positively correlated to logit l_y (denote as $p_y \propto l_y$ hereafter). According to the very foundation of statistic machine learning theory, the probability p_y here approximates a posterior $P(y|ob)$, where ob is the raw sample data so-called "observation". On the other hand, feature x is inferred via non-linear mapping f with ob, namely $x = f(ob)$. So feature x is well-known as a "representation" extracted from raw data ob to represent it. And w is the category's template vector. As an intuitive conclusion, the relational expression below could be written based on famous Bayes' Theorem:

$$P(y|ob) = P(y|x) = \frac{P(x|y)P(y)}{P(x)} \approx p_y \quad \propto \quad l_y = \|w_y\| \|x\| cos\theta_y \tag{4}$$

Carefully comparing each pair of the three terms on the left or right sides in (3) and (4), qualitative analysis and detailed suggestions could be drawn as follows:

The yth category's template vector w_y, is obtained either based on training samples of D_{train} belonging to category y as in supervised classification learning, or a statistic of the K-sized support set as in a C-way K-shot deployment. Therefore, w_y is independent of x, but only depends on the category y's distribution itself, just like a priori $P(y)$. Absolutely the same case for feature x, which contributes here its own a priori $P(x)$. Since $\cos\theta_y$ depends on both the category y plus the query feature x. Relevance between y and x would be formulated as a conditional probability $P(x|y)$.

The $\cos\theta_y$ is the common component for both inner-product or cosine logit prediction manners. It is worth noticing that as a query feature x closer to the category's basis w_y, the higher probability a sample x or ob is generated from the category y. Consequently, $\cos\theta_y$ is positively correlated to $P(x|y)$.

In addition, $\|x\|$ can be treated as an indicator of sample "quality". Low-quality samples that have the potential to belong to multiple categories would be assigned lower norm. Examples can be found in Fig. 4. Obviously, low-quality samples with the lower norm are indeed more likely to be randomly generated, while high-quality data which needs more specific details are on the contrary less likely to be sampled, thus $\|x\| \propto 1/P(x)$.

In Guo and al [16] introduced that L2 norm of classification weight vectors are inversely proportional to the number of training samples in every category. In our opinion, the scale of weight norm here stands for the diversity of the corresponding categories. To make it simple, we did a toy experiment and the result is illustrated in Fig. 3. Conversely speaking, the high diversity category which supports a larger subspace in the feature space, has a larger a priori $P(y)$ to be sampled, for whom higher $\|w_y\|$ would be obtained. Therefore, $\|w_y\| \propto P(y)$.

To summarize, for the inner-product logit prediction, three terms $\|w_y\|$, $\|x\|$ and $\cos\theta_y$ have their own hidden correspondence to $P(y)$, $P(x)$ and $P(x|y)$ in Bayes' Theorem respectively. In our method, l_{AFSL} eliminates the $\|w_y\|$ term compared to the inner-product fashion. The reason behind is that $\|w_y\|$, or $P(y)$ estimated from either the training set or support set is more likely to be biased. It is hard to say that the support set here can fully represent the category's distribution $P(y)$ with only limited samples in few-shot tasks. Removing the $\|w_y\|$ is equivalent to assuming that all categories follow an even distribution. As shown as the red dotted line in Fig. 3, the weight norm $\|w_y\|$ becomes much more the same even for unbalanced categories utilizing the assumption in training, which proves its effectiveness.

Differ from cosine method: Comparing our method (2) with the cosine method (1), the feature norm $\|x\|$ is preserved in l_{AFSL}. A simple explanation is that quality difference between samples is an objective existence, and hence $P(x)$ is an unbiased estimation that should not be canceled out. Since $\|x\|$ is automatically adjustable during training, there is no need for additional re-scale factor s anymore. At the same time, it can also filter the hard one in the samples through the weight assigned by $\|x\|$.

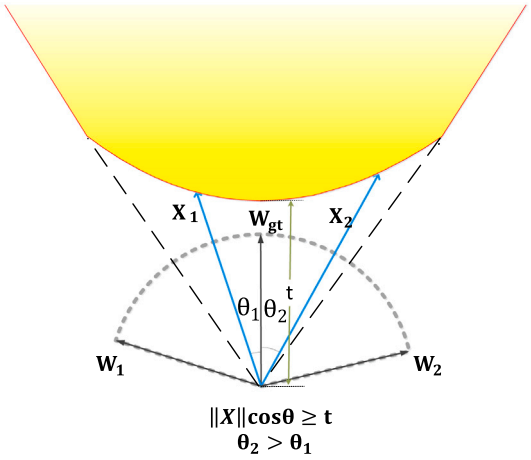


Fig. 2. The decision plan of AFSL metric function.

With Fig. 2, we may be able to understand it more easily. The red curve represents a decision surface, for samples' feature x dropping in the inner region with yellow mask will be recognized as class gt. It is a fan-shaped sector with a round head. Therefore, giving a threshold t, the larger angle of deviation θ_2 is tolerant for larger $\|x\|$, and more restricted θ_1 is required for smaller $\|x\|$, namely low-quality examples. This mechanism can be regarded as hard-sample-mining.

In summary, using this new logit prediction method can eliminate the biased a prior information caused by data imbalance in the training set, and taking sample quality into account in the decision procedure.

3.2. Extend the negative samples

As mentioned before. The hard sample can bring more information to the model. Increasing the number of negative samples in a batch can also increase the probability of hard samples. Recalling the episodic training algorithm, $m = K_s * C$ support samples and $n = K_q * C$ query samples are randomly selected from the training dataset. After that, network parameters are updated using the InfoNCE [17] loss as

$$Loss = \frac{1}{n}\sum_{i}\left[-d(x_i, c_{y_i}) + \log\sum_{k\in\hat{y}_i}\exp\left(d(x_i, c_k)\right)\right] \tag{5}$$

where $d(.)$ the logit prediction function. And c_y the prototype of class y which is normally mean of K_s support samples' feature vectors with ground-truth label y. Naturally, we adapt our l_{AFSL} in (2) as the logit prediction function $d(.)$ here. Consider an extreme setting, through replacing the prototype with each example independently, we should get the metric constraint which fully utilizes the contrastive information within one batch.

$$Loss_m = \frac{1}{n}\sum_{i}\left[-\frac{x_s^T x_i}{\|x_s\|} + \log\sum_{y_k\neq y_i}\exp\left(\frac{x_k^T x_i}{\|x_k\|}\right)\right] \tag{6}$$

Here x_s is the feature vector of support (positive) example sampled from the same category, while x_k the feature vector of any negative example. Since the loss function (6) is normally implemented by cross-entropy combining softmax and negative log-likelihood, it could be easily achieved by organizing the two examples from the same category side by side in a batch, and setting the diagonal of logit matrix to zero. The formula above can be regarded as the similarity logit between positive pairs should be larger than any of that between negative pairs, or that of the most-violated negative pair.

4. Experiments

4.1. Evaluation on open datasets

The proposed Adaptive Few-Shot Learning approach AFSL is evaluated on a series of testbeds, which are the general benchmark of few-shot learning, which are the general benchmark of few-shot learning. The mini-Imagenet dataset is a subset of ImageNet which is the most widely used benchmark for few-shot learning. There are 100 categories in the dataset, and 600 images in each category sum up to total 60,000 images. Following a common split, we use 64 categories for training, 16 and 20 categories for validation and testing respectively. The CUB200 dataset contains 11,788 bird images belonging to 200 species. A 100/50/50 partition of the training/validation/testing set is utilized here. The tiered-ImageNet is another larger subset of ILSVRC-12 with 608 classes and 779,165 images. Different from mini-Imagenet, the 608 classes here are grouped into 34 high-level nodes in the word-net hierarchy,

Table 1
Average accuracy with variance on tiered-Imagenet.

Approach	Network	1-shot	5-shots
protonet [5]	Conv-4	48.97 ± 0.21	66.47 ± 0.21
reptile [2]	Conv-4	53.31 ± 0.89	**72.69 ± 0.69**
simpleshot [12]	Conv-4	51.02 ± 0.20	68.98 ± 0.18
AFSL	Conv-4	**53.80 ± 0.09**	68.10 ± 0.08
LEO [20]	WRN	**66.33 ± 0.05**	**81.44 ± 0.09**
simpleshot [12]	WRN	63.85 ± 0.21	84.17 ± 0.15
AFSL	WRN	66.17 ± 0.1	79.79 ± 0.79

Table 2
Average accuracy with variance on CUB.

Approach	Network	1-shot	5-shots
matchingnet [4]	Conv-4	61.16 ± 0.89	72.86 ± 0.70
MAML [1]	Conv-4	54.73 ± 0.97	75.75 ± 0.76
baseline++[11]	Conv-4	60.53 ± 0.83	79.34 ± 0.61
protonet [5]	Conv-4	51.31 ± 0.91	70.77 ± 0.69
relationnet [21]	Conv-4	62.45 ± 0.98	77.84 ± 0.68
AFSL	Conv-4	**65.91 ± 0.10**	**81.09 ± 0.69**
baseline++[11]	ResNet-18	67.02 ± 0.90	83.58 ± 0.54
MAML [1]	ResNet-18	68.42 ± 1.07	83.47 ± 0.62
relationnet [21]	ResNet-18	68.58 ± 0.94	84.05 ± 0.56
protonet [5]	ResNet-18	72.99 ± 0.88	**86.64 ± 0.51**
matchingnet [4]	ResNet-18	73.49 ± 0.89	84.45 ± 0.58
AFSL	ResNet-18	**74.51 ± 0.99**	85.06 ± 0.65

then 20, 6 and 8 disjoint nodes are selected for training, validation and testing. This kind of split results in a more challenging, yet realistic regime where test classes are less similar to training/validation classes. For all images from any of the three datasets, we resize them into 84 × 84 pixels to fit the network via rescaling and center cropping.

We evaluate our methods using three different CNN's architectures:

- Four-layer Convolution Network (Conv-4): we follow the matchingnet [4] and protonet [5] to implement this baseline model;
- Residual Network (Res-18) [18]: a standard 18-layer architecture is adopted here, but the first two down-sampling layers are removed, and we change the first convolutional layer into a kernel size of 3 × 3.
- Wide Residual Network (WRN) [19]: we evaluate the WRN architecture with 28 convolutional layers and a widening factor of 10, namely WRN-28-10.

AFSL models are trained with different backbones. The Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$ is utilized during training, each episode includes 60-way and 1-shot samples, which is the upper bound of a batch size of our GPU devices. Data augmentation strategies follow as [5]. The cosine annealing learning rate schedule is performed with an initial setting of 0.001, and the cosine period is 80 epochs, with a total of 400 epochs for the entire training.

We measure the accuracy by drawing 5-way k-shots tasks from unseen categories in the test set, where $k = 1$ or 5 are noted as 1-shot or 5-shots. The higher scores are the better. The top accuracy of each track is printed in bold font, and each second-place one is underlined. 15 query images are sampled for test in each task, where 3 images from each category. This kind of few-shot task was repeated 500 times, so overall 7500 queries were tried for a randomly picked support set. Means and variances of the precision of AFSL and comparative methods are presented in Tables 1–3. It is a common practice to use the classification accuracy of 1-shot and 5-shots as the evaluation index of the task of few-shot classification. When using 1-shot, is similar to the verification task, while when using 5-shots, it is similar to the retrieval task. The former is obviously more difficult, because it is random to use a picture as the prototype in 1-shot, k shots for k > 1 will be more stable as the prototype because the mean value of multiple pictures is sampled. Compared with various prior works, the proposed AFSL achieves few-shot results that are better than or competitive to the state-of-the-art. It wins the top accuracy in half of the benchmarks, and in most of the rest it becomes the runner-up.

Among all the comparative methods, FEAT and simpleshot are close competitors. In FEAT, the authors introduced an attention mechanism and a transformer layer was added right after feature extraction, which assigns different weights to each category in the support set, so that it enables the neural network to better adapt to novel tasks. The simpleshot also improves the measurement of similarity, but focuses on improving the Euclidean distance, it uses the way of subtracting the mean and L2 normalizing center point's value, and has obtained excellent results. Together with our method, this way shows the importance of the measurement method. It can produce a good few-shot task result without complex network design. The FEAT also get a good performance but additional network structure and parameters were introduced in and hence they increased the model complexity.

4.2. Validation study

In order to verify the impact of the dataset on the model, including the distribution of data categories will affect the weight of classification in the model and the sample's quality will affect the feature's norm value. We designed such an experiment: we

Table 3
Average accuracy with variance on mini-Imagenet.

Approach	Network	1-shot	5-shots
matchingnet [4]	Conv-4	43.56 ± 0.84	55.31 ± 0.73
MAML [1]	Conv-4	48.70 ± 1.84	63.11 ± 0.94
protonet [5]	Conv-4	49.97 ± 0.78	68.2 ± 0.66
simpleshot [12]	Conv-4	49.69 ± 0.19	66.92 ± 0.17
reptile [2]	Conv-4	49.42 ± 0.32	65.99 ± 0.58
FGNN [6]	Conv-4	50.33 ± 0.36	66.41 ± 0.63
relationnet [21]	Conv-4	50.44 ± 0.82	65.32 ± 0.70
Qiao and al[22]	Conv-4	54.53 ± 0.40	67.87 ± 0.20
FEAT [7]	Conv-4	**55.15 ± 0.20**	**71.61 ± 0.16**
AFSL	Conv-4	53.88 ± 0.90	69.08 ± 0.76
MAML [1]	ResNet-18	49.61 ± 0.92	65.72 ± 0.77
baseline++[11]	ResNet-18	51.87 ± 0.77	75.68 ± 0.63
relationnet [21]	ResNet-18	52.48 ± 0.86	69.83 ± 0.68
matchingnet [4]	ResNet-18	52.91 ± 0.88	68.88 ± 0.69
protonet [5]	ResNet-18	54.16 ± 0.82	73.68 ± 0.65
simpleshot [12]	ResNet-18	63.50 ± 0.20	**80.33 ± 0.14**
AFSL	ResNet-18	**63.77 ± 0.90**	79.44 ± 0.61
Qiao and al[22]	WRN	59.60 ± 0.41	73.74 ± 0.19
matchingnet [4]	WRN	64.03 ± 0.20	76.32 ± 0.16
protonet [5]	WRN	62.60 ± 0.20	79.97 ± 0.14
LEO [20]	WRN	61.76 ± 0.08	77.59 ± 0.12
simpleshot [12]	WRN	62.85 ± 0.20	80.02 ± 0.14
FEAT [7]	WRN	65.10 ± 0.20	**81.11 ± 0.14**
AFSL	WRN	**65.88 ± 0.92**	80.12 ± 0.64

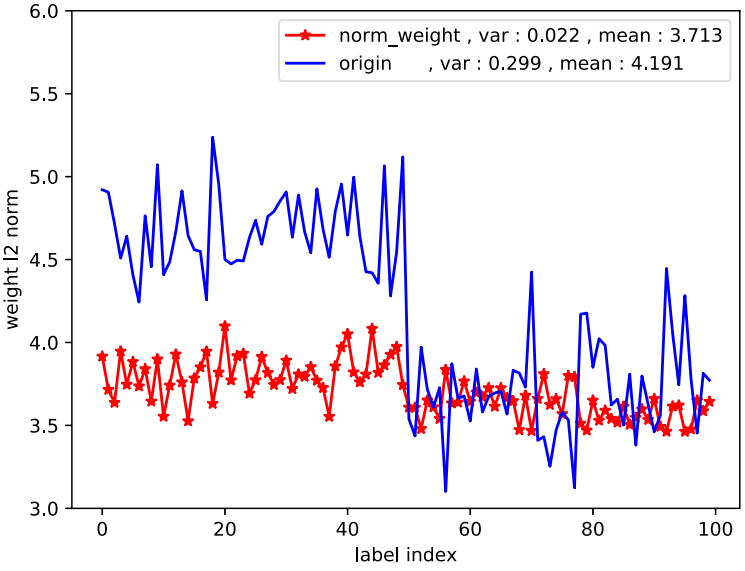


Fig. 3. The comparison of category-wise weight L2 norm between original method and AFSL.

use the Conv-4 mentioned above as backbone, add a full-connect layer for 100 categories of classification tasks on mini-Imagenet dataset.

Models are trained in an "unbalanced" fashion, where 200 images are included in each of the former 50 categories, and for the latter 50 categories, it only consists of 10 images each. The data comes from mini-Imagenet. The original cosine metric and AFSL metric are used respectively, over 400 epochs, we recorded the weight values of the full-connect layer's L2 norm of the weight. We get the results shown in Fig. 3. The horizontal axis represents the corresponding category index and the vertical axis represents values of weight L2 norm. The blue line denotes the weight L2 norm values of the original method, and the red dotted line represents that of our AFSL (better viewed in color). The weight L2 norm of categories including more samples are obviously larger than the opposite ones for the original method, and this situation is significantly reduced for AFSL. A conclusion can be intuitively drawn that a weight normalization adopted in AFSL should suppress the negative impact caused by data imbalance. Then, some images are sampled from the dataset, the AFSL model obtained above is used for forwarding propagation to get the feature vector, and their norms are calculated. The corresponding the relationship between image's quality and the norm of feature is shown in Fig. 4.



Fig. 4. Sampled images and corresponding L2 norm of feature vectors.

Table 4
Results of ablation experiment on CUB200 dataset.

Metric func	w/o prototype	1-shot	Growth		5-shots	Growth	
			Relative	Overall		Relative	Overall
cosine	✗	61.3	–	–	76.6	–	–
	✓	61.9	0.98%	0.98%	77.1	0.65%	0.65%
inner	✗	62.8	–	2.44%	77.8	–	1.56%
	✓	63.7	1.43%	3.91%	78.0	0.25%	1.82%
AFSL	✗	64.1	–	4.57%	79.1	–	3.26%
	✓	65.8	2.65%	7.50%	81.0	2.40%	5.74%

Here eight images sampled from mini-Imagenet are exhibited. The first line of texts under the images represents the corresponding category of each image, and the numbers below are their L2 norm of feature vectors. Obviously low-normed images are either tiny, blurred, or pale. On the contrary, the image corresponding to the value of high norm is clear, bright, and prominent. The above contents can well prove that the norm value of image features is indeed related to image quality.

4.3. Ablation study

We also perform an ablation experiment to reveal the effectiveness of the improvements we made. Specifically, we train the Conv-4 network on CUB dataset, in cases the main techniques are separately switched on or off. The results of controlled experiments are shown in Table 4. The 1-shot & 5-shots of the accuracy is measured over 1000 rounds of randomly assembled 5-way tasks. The *cosine* method uses cosine logit function with re-scale factor $s = 32$. The *inner* method uses inner logit function, as described in Section 3.1. The *AFSL* method uses our AFSL logit function. The *w/o prototype* means whether or not to use "prototypes". The cross mark indicates the original way as in protonet [5], which is treated as the baseline, and the tick mark indicates "without prototypes", that is our way to extend negative samples. The *growth* contains two columns, *overall* represents the percentage increase in accuracy of the current method compared with base cosine method. While *relative* represents the percentage increase of each method compared with each base method. The items which compare with themselves are represented by "-".

According to the *growth* columns in Table 4, we can draw the following conclusions. Firstly, the AFSL obtained the best results, the inner-product method is between the two, and the cosine method is the worst. Secondly, extending negative samples is an effective improvement. It can be observed that the three comparison methods have been improved. Finally, the effectiveness of extended negative samples method is more significant when combining it with logit prediction than it working alone, especially with AFSL. We analyzed and try to explain it as follows. As mentioned in Section 3.1, The inner method can also play a relevant role as AFSL, but the training data have a relatively negative impact on it. Our logit prediction function has effects like hard example mining and dirty label filtering compared to cosine method. Only along with the increase in negative samples at one batch, the model did not get much obvious benefit, it does not really focus on the hard samples. So the promotion of accuracy when two strategies were both activated is not a unilateral impact of extended negative samples. Instead, it helps the logit prediction to perform better in that case.

5. Conclusion

Our method proves once again that the improvement in a metric way has great potential to develop the performance of the model. In the validation study, the designed experiments verify our proposal that relationship between data distribution and model, as well as sample quality and feature norm. And it also intuitively prove the improvement effect of our method. In the ablation study, compared with the original method, our method achieved 7.5% and 5.74% improvement on 1-shot and 5-shots respectively, they are better than 3.91% and 1.82% in inner-product way. The above contents show the rationality and effectiveness of our improved method. And its success also can be transplanted, other few-shot classification task frameworks based on metric function are worth trying our method. However, we have tried to add some marginal parameter to the metric function, but we have not achieved better performance. In the future, we will continue to dig into relevant principles and try to add the large-margin constraints into our framework.

CRediT authorship contribution statement

Xinke Zeng: Methodology, Software, Data curation, Writing – original draft. **Bo Huang:** Conceptualization, Supervision, Resources, Writing – original draft. **Ke Jia:** Methodology, Funding acquisition, Writing – review & editing. **Li Jia:** Validation, Formal analysis. **Ke Zhao:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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